```
#checking for GPU
import tensorflow as tf
print("GPU Available:", tf.config.list_physical_devices('GPU'))
```

```
GPU Available: [PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU
```

```
!ls
```

```
sample_data
```

```
!pip install -q kaggle
```

```
import json

kaggle_data = {
    "username": "Kaggle",
    "key": "KGAT_94badf1021e997ffa82a3299b41bd531"
}

with open("kaggle.json", "w") as f:
    json.dump(kaggle_data, f)

print("kaggle.json file created!")
```

```
kaggle.json file created!
```

```
!pip install -q kaggle
!mkdir -p ~/.kaggle
!cp kaggle.json ~/.kaggle/
!chmod 600 ~/.kaggle/kaggle.json
```

```
!kaggle datasets download -d masoudnickparvar/brain-tumor-mri-dataset
!unzip brain-tumor-mri-dataset.zip
```

```
⚠[1;30;43mStreaming output truncated to the last 5000 lines.⚠[0m
  inflating: Training/glioma/Tr-gl_279.jpg
  inflating: Training/glioma/Tr-gl_28.jpg
  inflating: Training/glioma/Tr-gl_280.jpg
  inflating: Training/glioma/Tr-gl_281.jpg
  inflating: Training/glioma/Tr-gl_282.jpg
  inflating: Training/glioma/Tr-gl_283.jpg
  inflating: Training/glioma/Tr-gl_284.jpg
  inflating: Training/glioma/Tr-gl_285.jpg
  inflating: Training/glioma/Tr-gl_286.jpg
```

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inflating: Training/glioma/Tr-gl_289.jpg
inflating: Training/glioma/Tr-gl_29.jpg
inflating: Training/glioma/Tr-gl_290.jpg
inflating: Training/glioma/Tr-gl_291.jpg
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inflating: Training/glioma/Tr-gl_30.jpg
inflating: Training/glioma/Tr-gl_300.jpg
inflating: Training/glioma/Tr-gl_301.jpg
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inflating: Training/glioma/Tr-gl_309.jpg
inflating: Training/glioma/Tr-gl_31.jpg
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inflating: Training/glioma/Tr-gl_414.jpg
inflating: Training/glioma/Tr-gl_415.jpg
```

```
  inflating: Training/pituitary/Tr-pi_957.jpg
  inflating: Training/pituitary/Tr-pi_958.jpg
  inflating: Training/pituitary/Tr-pi_959.jpg
  inflating: Training/pituitary/Tr-pi_96.jpg
  inflating: Training/pituitary/Tr-pi_960.jpg
  inflating: Training/pituitary/Tr-pi_961.jpg
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  inflating: Training/pituitary/Tr-pi_998.jpg
  inflating: Training/pituitary/Tr-pi_999.jpg
```

```
!ls
```

```
brain-tumor-mri-dataset.zip  kaggle.json  sample_data  Testing  Training

## connecting data set properly
train_path = "/content/Training"
test_path = "/content/Testing"

import os
print("Classes:", os.listdir(train_path))

Classes: ['pituitary', 'meningioma', 'glioma', 'notumor']

# create data generator
from tensorflow.keras.preprocessing.image import ImageDataGenerator

train_datagen = ImageDataGenerator(
    rescale=1./255,
    rotation_range=15,
    zoom_range=0.1,
    width_shift_range=0.1,
    height_shift_range=0.1,
    horizontal_flip=True
)

test_datagen = ImageDataGenerator(rescale=1./255)

train_generator = train_datagen.flow_from_directory(
    train_path,
    target_size=(224,224),
    batch_size=32,
    class_mode='categorical'
)

test_generator = test_datagen.flow_from_directory(
    test_path,
    target_size=(224,224),
    batch_size=32,
    class_mode='categorical',
    shuffle=False
)

Found 5600 images belonging to 4 classes.
Found 1600 images belonging to 4 classes.

## Build MobileVetNetV2 Model - it is Pretrained CNN model
import tensorflow as tf
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.layers import Dense, GlobalAveragePooling2D, Dropout
from tensorflow.keras.models import Model
```

```
base_model = MobileNetV2(
    weights='imagenet',
    include_top=False,
    input_shape=(224,224,3)
)

base_model.trainable = False

x = base_model.output
x = GlobalAveragePooling2D()(x)
x = Dropout(0.5)(x)
output = Dense(4, activation='softmax')(x)

model = Model(inputs=base_model.input, outputs=output)

model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

model.summary()
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applicat
◆[1m9406464/9406464◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m
```

Model: "functional"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 224, 224, 3)	0	-
Conv1 (Conv2D)	(None, 112, 112, 32)	864	input_layer[0][0]
bn_Conv1 (BatchNormalizatio…	(None, 112, 112, 32)	128	Conv1[0][0]
Conv1_relu (ReLU)	(None, 112, 112, 32)	0	bn_Conv1[0][0]
expanded_conv_dept… (DepthwiseConv2D)	(None, 112, 112, 32)	288	Conv1_relu[0][0]
expanded_conv_dept… (BatchNormalizatio…	(None, 112, 112, 32)	128	expanded_conv_de…

expanded_conv_dept… (ReLU)	(None, 112, 112, 32)	0	expanded_conv_de…
expanded_conv_proj… (Conv2D)	(None, 112, 112, 16)	512	expanded_conv_de…
expanded_conv_proj… (BatchNormalizatio…	(None, 112, 112, 16)	64	expanded_conv_pr…
block_1_expand (Conv2D)	(None, 112, 112, 96)	1,536	expanded_conv_pr…
block_1_expand_BN (BatchNormalizatio…	(None, 112, 112, 96)	384	block_1_expand[0…
block_1_expand_relu (ReLU)	(None, 112, 112, 96)	0	block_1_expand_B…
block_1_pad (ZeroPadding2D)	(None, 113, 113, 96)	0	block_1_expand_r…
block_1_depthwise (DepthwiseConv2D)	(None, 56, 56, 96)	864	block_1_pad[0][0]
block_1_depthwise_… (BatchNormalizatio…	(None, 56, 56, 96)	384	block_1_depthwis…
block_1_depthwise_… (ReLU)	(None, 56, 56, 96)	0	block_1_depthwis…
block_1_project (Conv2D)	(None, 56, 56, 24)	2,304	block_1_depthwis…
block_1_project_BN (BatchNormalizatio…	(None, 56, 56, 24)	96	block_1_project[…
block_2_expand (Conv2D)	(None, 56, 56, 144)	3,456	block_1_project_…
block_2_expand_BN (BatchNormalizatio…	(None, 56, 56, 144)	576	block_2_expand[0…
block_2_expand_relu (ReLU)	(None, 56, 56, 144)	0	block_2_expand_B…
block_2_depthwise (DepthwiseConv2D)	(None, 56, 56, 144)	1,296	block_2_expand_r…

block_2_depthwise_… (BatchNormalizatio…	(None, 56, 56, 144)	576	block_2_depthwis…
block_2_depthwise_… (ReLU)	(None, 56, 56, 144)	0	block_2_depthwis…
block_2_project (Conv2D)	(None, 56, 56, 24)	3,456	block_2_depthwis…
block_2_project_BN (BatchNormalizatio…	(None, 56, 56, 24)	96	block_2_project[…
block_2_add (Add)	(None, 56, 56, 24)	0	block_1_project_… block_2_project_…
block_3_expand (Conv2D)	(None, 56, 56, 144)	3,456	block_2_add[0][0]
block_3_expand_BN (BatchNormalizatio…	(None, 56, 56, 144)	576	block_3_expand[0…
block_3_expand_relu (ReLU)	(None, 56, 56, 144)	0	block_3_expand_B…
block_3_pad (ZeroPadding2D)	(None, 57, 57, 144)	0	block_3_expand_r…
block_3_depthwise (DepthwiseConv2D)	(None, 28, 28, 144)	1,296	block_3_pad[0][0]
block_3_depthwise_… (BatchNormalizatio…	(None, 28, 28, 144)	576	block_3_depthwis…
block_3_depthwise_… (ReLU)	(None, 28, 28, 144)	0	block_3_depthwis…
block_3_project (Conv2D)	(None, 28, 28, 32)	4,608	block_3_depthwis…
block_3_project_BN (BatchNormalizatio…	(None, 28, 28, 32)	128	block_3_project[…
block_4_expand (Conv2D)	(None, 28, 28, 192)	6,144	block_3_project_…
block_4_expand_BN (BatchNormalizatio…	(None, 28, 28, 192)	768	block_4_expand[0…

block_4_expand_relu (ReLU)	(None, 28, 28, 192)	0	block_4_expand_B…
block_4_depthwise (DepthwiseConv2D)	(None, 28, 28, 192)	1,728	block_4_expand_r…
block_4_depthwise_… (BatchNormalizatio…	(None, 28, 28, 192)	768	block_4_depthwis…
block_4_depthwise_… (ReLU)	(None, 28, 28, 192)	0	block_4_depthwis…
block_4_project (Conv2D)	(None, 28, 28, 32)	6,144	block_4_depthwis…
block_4_project_BN (BatchNormalizatio…	(None, 28, 28, 32)	128	block_4_project[…
block_4_add (Add)	(None, 28, 28, 32)	0	block_3_project_… block_4_project_…
block_5_expand (Conv2D)	(None, 28, 28, 192)	6,144	block_4_add[0][0]
block_5_expand_BN (BatchNormalizatio…	(None, 28, 28, 192)	768	block_5_expand[0…
block_5_expand_relu (ReLU)	(None, 28, 28, 192)	0	block_5_expand_B…
block_5_depthwise (DepthwiseConv2D)	(None, 28, 28, 192)	1,728	block_5_expand_r…
block_5_depthwise_… (BatchNormalizatio…	(None, 28, 28, 192)	768	block_5_depthwis…
block_5_depthwise_… (ReLU)	(None, 28, 28, 192)	0	block_5_depthwis…
block_5_project (Conv2D)	(None, 28, 28, 32)	6,144	block_5_depthwis…
block_5_project_BN (BatchNormalizatio…	(None, 28, 28, 32)	128	block_5_project[…
block_5_add (Add)	(None, 28, 28, 32)	0	block_4_add[0][0… block_5_project_…

block_6_expand (Conv2D)	(None, 28, 28, 192)	6,144	block_5_add[0][0]
block_6_expand_BN (BatchNormalizatio…	(None, 28, 28, 192)	768	block_6_expand[0…
block_6_expand_relu (ReLU)	(None, 28, 28, 192)	0	block_6_expand_B…
block_6_pad (ZeroPadding2D)	(None, 29, 29, 192)	0	block_6_expand_r…
block_6_depthwise (DepthwiseConv2D)	(None, 14, 14, 192)	1,728	block_6_pad[0][0]
block_6_depthwise_… (BatchNormalizatio…	(None, 14, 14, 192)	768	block_6_depthwis…
block_6_depthwise_… (ReLU)	(None, 14, 14, 192)	0	block_6_depthwis…
block_6_project (Conv2D)	(None, 14, 14, 64)	12,288	block_6_depthwis…
block_6_project_BN (BatchNormalizatio…	(None, 14, 14, 64)	256	block_6_project[…
block_7_expand (Conv2D)	(None, 14, 14, 384)	24,576	block_6_project_…
block_7_expand_BN (BatchNormalizatio…	(None, 14, 14, 384)	1,536	block_7_expand[0…
block_7_expand_relu (ReLU)	(None, 14, 14, 384)	0	block_7_expand_B…
block_7_depthwise (DepthwiseConv2D)	(None, 14, 14, 384)	3,456	block_7_expand_r…
block_7_depthwise_… (BatchNormalizatio…	(None, 14, 14, 384)	1,536	block_7_depthwis…
block_7_depthwise_… (ReLU)	(None, 14, 14, 384)	0	block_7_depthwis…
block_7_project (Conv2D)	(None, 14, 14, 64)	24,576	block_7_depthwis…

block_7_project_BN (BatchNormalizatio…	(None, 14, 14, 64)	256	block_7_project[…
block_7_add (Add)	(None, 14, 14, 64)	0	block_6_project_… block_7_project_…
block_8_expand (Conv2D)	(None, 14, 14, 384)	24,576	block_7_add[0][0]
block_8_expand_BN (BatchNormalizatio…	(None, 14, 14, 384)	1,536	block_8_expand[0…
block_8_expand_relu (ReLU)	(None, 14, 14, 384)	0	block_8_expand_B…
block_8_depthwise (DepthwiseConv2D)	(None, 14, 14, 384)	3,456	block_8_expand_r…
block_8_depthwise_… (BatchNormalizatio…	(None, 14, 14, 384)	1,536	block_8_depthwis…
block_8_depthwise_… (ReLU)	(None, 14, 14, 384)	0	block_8_depthwis…
block_8_project (Conv2D)	(None, 14, 14, 64)	24,576	block_8_depthwis…
block_8_project_BN (BatchNormalizatio…	(None, 14, 14, 64)	256	block_8_project[…
block_8_add (Add)	(None, 14, 14, 64)	0	block_7_add[0][0… block_8_project_…
block_9_expand (Conv2D)	(None, 14, 14, 384)	24,576	block_8_add[0][0]
block_9_expand_BN (BatchNormalizatio…	(None, 14, 14, 384)	1,536	block_9_expand[0…
block_9_expand_relu (ReLU)	(None, 14, 14, 384)	0	block_9_expand_B…
block_9_depthwise (DepthwiseConv2D)	(None, 14, 14, 384)	3,456	block_9_expand_r…
block_9_depthwise_… (BatchNormalizatio…	(None, 14, 14, 384)	1,536	block_9_depthwis…

block_9_depthwise_… (ReLU)	(None, 14, 14, 384)	0	block_9_depthwis…
block_9_project (Conv2D)	(None, 14, 14, 64)	24,576	block_9_depthwis…
block_9_project_BN (BatchNormalizatio…	(None, 14, 14, 64)	256	block_9_project[…
block_9_add (Add)	(None, 14, 14, 64)	0	block_8_add[0][0… block_9_project_…
block_10_expand (Conv2D)	(None, 14, 14, 384)	24,576	block_9_add[0][0]
block_10_expand_BN (BatchNormalizatio…	(None, 14, 14, 384)	1,536	block_10_expand[…
block_10_expand_re… (ReLU)	(None, 14, 14, 384)	0	block_10_expand_…
block_10_depthwise (DepthwiseConv2D)	(None, 14, 14, 384)	3,456	block_10_expand_…
block_10_depthwise… (BatchNormalizatio…	(None, 14, 14, 384)	1,536	block_10_depthwi…
block_10_depthwise… (ReLU)	(None, 14, 14, 384)	0	block_10_depthwi…
block_10_project (Conv2D)	(None, 14, 14, 96)	36,864	block_10_depthwi…
block_10_project_BN (BatchNormalizatio…	(None, 14, 14, 96)	384	block_10_project…
block_11_expand (Conv2D)	(None, 14, 14, 576)	55,296	block_10_project…
block_11_expand_BN (BatchNormalizatio…	(None, 14, 14, 576)	2,304	block_11_expand[…
block_11_expand_re… (ReLU)	(None, 14, 14, 576)	0	block_11_expand_…
block_11_depthwise (DepthwiseConv2D)	(None, 14, 14, 576)	5,184	block_11_expand_…

block_11_depthwise… (BatchNormalizatio…	(None, 14, 14, 576)	2,304	block_11_depthwi…
block_11_depthwise… (ReLU)	(None, 14, 14, 576)	0	block_11_depthwi…
block_11_project (Conv2D)	(None, 14, 14, 96)	55,296	block_11_depthwi…
block_11_project_BN (BatchNormalizatio…	(None, 14, 14, 96)	384	block_11_project…
block_11_add (Add)	(None, 14, 14, 96)	0	block_10_project… block_11_project…
block_12_expand (Conv2D)	(None, 14, 14, 576)	55,296	block_11_add[0][…
block_12_expand_BN (BatchNormalizatio…	(None, 14, 14, 576)	2,304	block_12_expand[…
block_12_expand_re… (ReLU)	(None, 14, 14, 576)	0	block_12_expand_…
block_12_depthwise (DepthwiseConv2D)	(None, 14, 14, 576)	5,184	block_12_expand_…
block_12_depthwise… (BatchNormalizatio…	(None, 14, 14, 576)	2,304	block_12_depthwi…
block_12_depthwise… (ReLU)	(None, 14, 14, 576)	0	block_12_depthwi…
block_12_project (Conv2D)	(None, 14, 14, 96)	55,296	block_12_depthwi…
block_12_project_BN (BatchNormalizatio…	(None, 14, 14, 96)	384	block_12_project…
block_12_add (Add)	(None, 14, 14, 96)	0	block_11_add[0][… block_12_project…
block_13_expand (Conv2D)	(None, 14, 14, 576)	55,296	block_12_add[0][…
block_13_expand_BN (BatchNormalizatio…	(None, 14, 14, 576)	2,304	block_13_expand[…

block_13_expand_re… (ReLU)	(None, 14, 14, 576)	0	block_13_expand_…
block_13_pad (ZeroPadding2D)	(None, 15, 15, 576)	0	block_13_expand_…
block_13_depthwise (DepthwiseConv2D)	(None, 7, 7, 576)	5,184	block_13_pad[0][…
block_13_depthwise… (BatchNormalizatio…	(None, 7, 7, 576)	2,304	block_13_depthwi…
block_13_depthwise… (ReLU)	(None, 7, 7, 576)	0	block_13_depthwi…
block_13_project (Conv2D)	(None, 7, 7, 160)	92,160	block_13_depthwi…
block_13_project_BN (BatchNormalizatio…	(None, 7, 7, 160)	640	block_13_project…
block_14_expand (Conv2D)	(None, 7, 7, 960)	153,600	block_13_project…
block_14_expand_BN (BatchNormalizatio…	(None, 7, 7, 960)	3,840	block_14_expand[…
block_14_expand_re… (ReLU)	(None, 7, 7, 960)	0	block_14_expand_…
block_14_depthwise (DepthwiseConv2D)	(None, 7, 7, 960)	8,640	block_14_expand_…
block_14_depthwise… (BatchNormalizatio…	(None, 7, 7, 960)	3,840	block_14_depthwi…
block_14_depthwise… (ReLU)	(None, 7, 7, 960)	0	block_14_depthwi…
block_14_project (Conv2D)	(None, 7, 7, 160)	153,600	block_14_depthwi…
block_14_project_BN (BatchNormalizatio…	(None, 7, 7, 160)	640	block_14_project…
block_14_add (Add)	(None, 7, 7, 160)	0	block_13_project… block_14_project…

block_15_expand (Conv2D)	(None, 7, 7, 960)	153,600	block_14_add[0][…
block_15_expand_BN (BatchNormalizatio…	(None, 7, 7, 960)	3,840	block_15_expand[…
block_15_expand_re… (ReLU)	(None, 7, 7, 960)	0	block_15_expand_…
block_15_depthwise (DepthwiseConv2D)	(None, 7, 7, 960)	8,640	block_15_expand_…
block_15_depthwise… (BatchNormalizatio…	(None, 7, 7, 960)	3,840	block_15_depthwi…
block_15_depthwise… (ReLU)	(None, 7, 7, 960)	0	block_15_depthwi…
block_15_project (Conv2D)	(None, 7, 7, 160)	153,600	block_15_depthwi…
block_15_project_BN (BatchNormalizatio…	(None, 7, 7, 160)	640	block_15_project…
block_15_add (Add)	(None, 7, 7, 160)	0	block_14_add[0][… block_15_project…
block_16_expand (Conv2D)	(None, 7, 7, 960)	153,600	block_15_add[0][…
block_16_expand_BN (BatchNormalizatio…	(None, 7, 7, 960)	3,840	block_16_expand[…
block_16_expand_re… (ReLU)	(None, 7, 7, 960)	0	block_16_expand_…
block_16_depthwise (DepthwiseConv2D)	(None, 7, 7, 960)	8,640	block_16_expand_…
block_16_depthwise… (BatchNormalizatio…	(None, 7, 7, 960)	3,840	block_16_depthwi…
block_16_depthwise… (ReLU)	(None, 7, 7, 960)	0	block_16_depthwi…
block_16_project (Conv2D)	(None, 7, 7, 320)	307,200	block_16_depthwi…

block_16_project_BN (BatchNormalizatio…	(None, 7, 7, 320)	1,280	block_16_project…
Conv_1 (Conv2D)	(None, 7, 7, 1280)	409,600	block_16_project…
Conv_1_bn (BatchNormalizatio…	(None, 7, 7, 1280)	5,120	Conv_1[0][0]
out_relu (ReLU)	(None, 7, 7, 1280)	0	Conv_1_bn[0][0]
global_average_poo… (GlobalAveragePool…	(None, 1280)	0	out_relu[0][0]
dropout (Dropout)	(None, 1280)	0	global_average_p…
dense (Dense)	(None, 4)	5,124	dropout[0][0]

Total params: 2,263,108 (8.63 MB)

Trainable params: 5,124 (20.02 KB)

Non-trainable params: 2,257,984 (8.61 MB)

```
## TRAIN THE MODEL
history = model.fit(
    train_generator,
    validation_data=test_generator,
    epochs=10
)
```

```
Epoch 1/10
◆[1m175/175◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m96s◆[0m 445ms/s
Epoch 2/10
◆[1m175/175◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m76s◆[0m 432ms/s
Epoch 3/10
◆[1m175/175◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m77s◆[0m 400ms/s
Epoch 4/10
◆[1m175/175◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m70s◆[0m 401ms/s
Epoch 5/10
◆[1m175/175◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m70s◆[0m 402ms/s
Epoch 6/10
◆[1m175/175◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m70s◆[0m 402ms/s
Epoch 7/10
◆[1m175/175◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m70s◆[0m 402ms/s
Epoch 8/10
◆[1m175/175◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m71s◆[0m 405ms/s
```

```
Epoch 9/10
◇[1m175/175◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m70s◇[0m 400ms/s
Epoch 10/10
◇[1m175/175◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m70s◇[0m 400ms/s

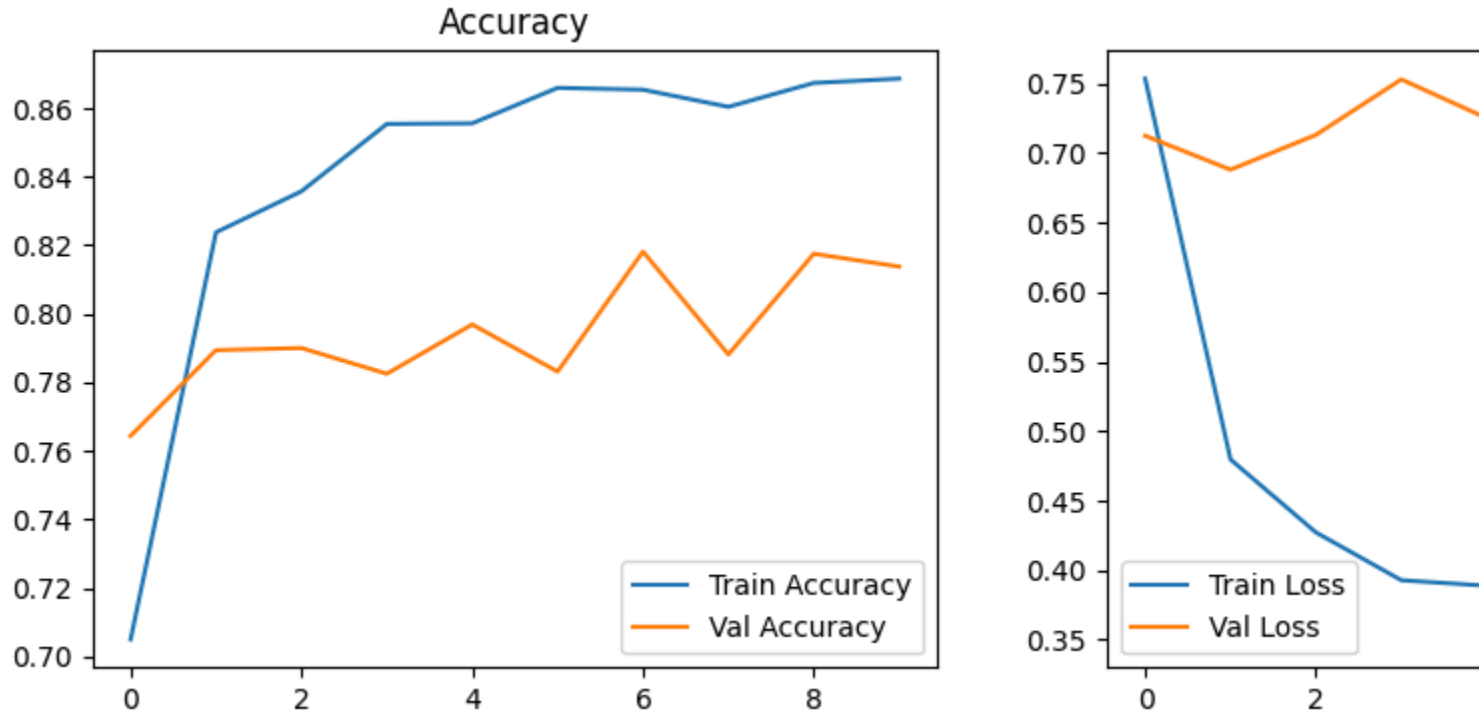
# Trainig graph
import matplotlib.pyplot as plt

plt.figure(figsize=(12,4))

plt.subplot(1,2,1)
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Val Accuracy')
plt.legend()
plt.title("Accuracy")

plt.subplot(1,2,2)
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Val Loss')
plt.legend()
plt.title("Loss")

plt.show()
```



png

```
# Confusion matrix and classification Report
import numpy as np
```

```
from sklearn.metrics import confusion_matrix, classification_report
import seaborn as sns
import matplotlib.pyplot as plt

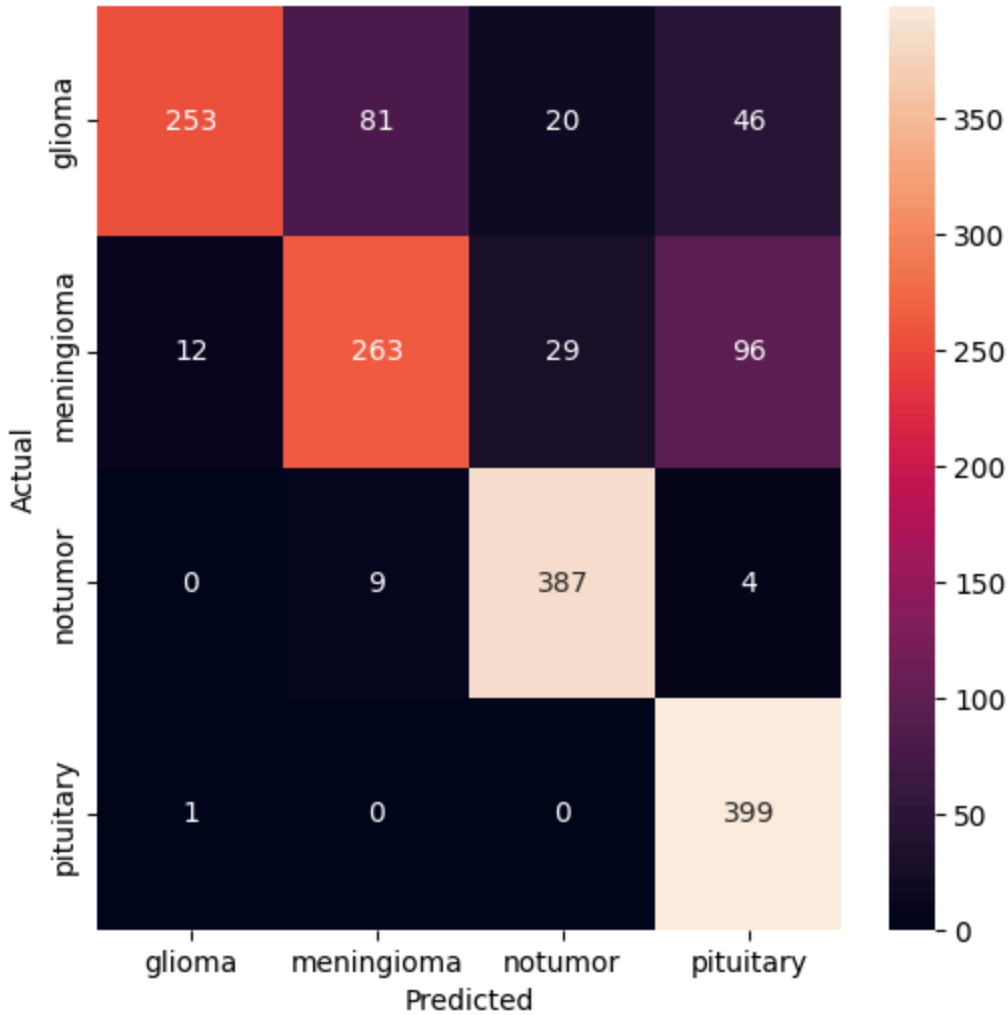
pred = model.predict(test_generator)
y_pred = np.argmax(pred, axis=1)

cm = confusion_matrix(test_generator.classes, y_pred)

plt.figure(figsize=(6,6))
sns.heatmap(cm, annot=True, fmt='d',
            xticklabels=train_generator.class_indices.keys(),
            yticklabels=train_generator.class_indices.keys())
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

print(classification_report(test_generator.classes, y_pred))
```

```
◇[1m50/50◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m8s◇[0m 70ms/step
```



png

```
              precision    recall  f1-score   support

           0       0.95      0.63      0.76       400
           1       0.75      0.66      0.70       400
           2       0.89      0.97      0.93       400
           3       0.73      1.00      0.84       400

    accuracy                           0.81      1600
   macro avg       0.83      0.81      0.81      1600
weighted avg       0.83      0.81      0.81      1600
```

```
# Visualizing sample prediction
class_names = list(train_generator.class_indices.keys())

images, labels = next(test_generator)

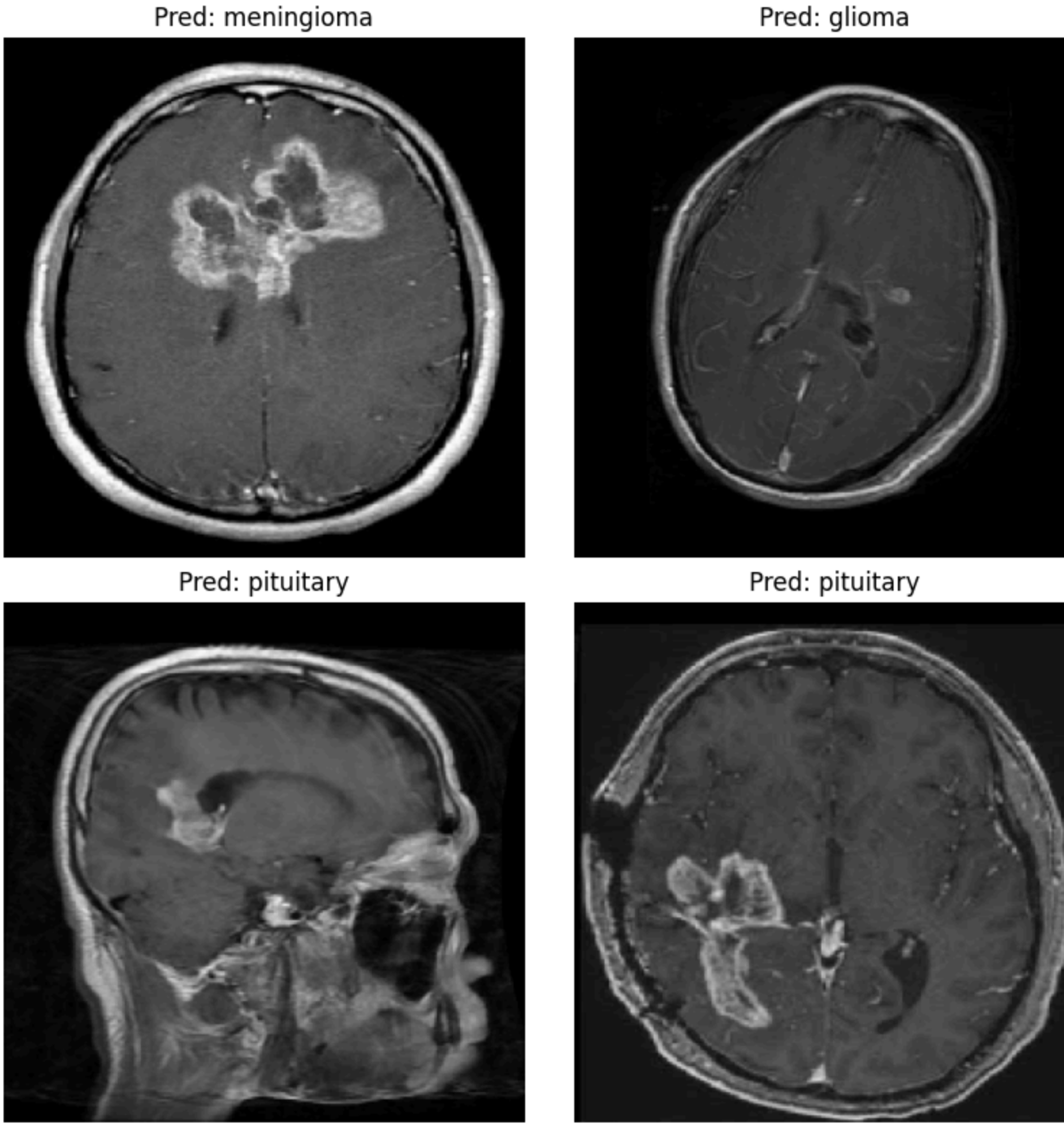
predictions = model.predict(images)
pred_classes = np.argmax(predictions, axis=1)
```

```
plt.figure(figsize=(12,8))

for i in range(6):
    plt.subplot(2,3,i+1)
    plt.imshow(images[i])
    plt.title(f"Pred: {class_names[pred_classes[i]]}")
    plt.axis("off")

plt.tight_layout()
plt.show()
```

```
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m4s◇[0m 4s/step
```



png

```
class_names = list(train_generator.class_indices.keys())

## Create Grad-CAM function
import cv2
```

```
def make_gradcam_heatmap(img_array, model, last_conv_layer_name):

    grad_model = tf.keras.models.Model(
        [model.inputs],
        [model.get_layer(last_conv_layer_name).output, model.output]
    )

    with tf.GradientTape() as tape:
        conv_outputs, predictions = grad_model(img_array)
        class_index = tf.argmax(predictions[0])
        loss = predictions[:, class_index]

    grads = tape.gradient(loss, conv_outputs)
    pooled_grads = tf.reduce_mean(grads, axis=(0,1,2))

    conv_outputs = conv_outputs[0]
    heatmap = conv_outputs @ pooled_grads[..., tf.newaxis]
    heatmap = tf.squeeze(heatmap)

    heatmap = np.maximum(heatmap, 0) / np.max(heatmap)

    return heatmap

os.listdir()

['.config',
 'brain-tumor-mri-dataset.zip',
 'Testing',
 'kaggle.json',
 'Training',
 'sample_data']

from tensorflow.keras.preprocessing import image
import numpy as np

img_path = test_generator.filepaths[0]

from tensorflow.keras.preprocessing import image
import numpy as np

# get image path
img_path = test_generator.filepaths[0]

# load image
img = image.load_img(img_path, target_size=(224,224))

# convert to array
img_array = image.img_to_array(img)
```

```
# normalize
img_array = img_array / 255.0

# add batch dimension
img_array = np.expand_dims(img_array, axis=0)

#Get Prediction
pred = model.predict(img_array)
pred_class = np.argmax(pred)
confidence = np.max(pred)

class_names = list(train_generator.class_indices.keys())
predicted_label = class_names[pred_class]

if predicted_label == "notumor":
    tumor_status = "No Tumor Detected"
else:
    tumor_status = "Tumor Detected"

◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m12s◇[0m 12s/step

# Apply Grad-CAM
last_conv_layer_name = "Conv_1"

heatmap = make_gradcam_heatmap(img_array, model, last_conv_layer_name)

heatmap = cv2.resize(heatmap, (224,224))
heatmap = np.uint8(255 * heatmap)
heatmap = cv2.applyColorMap(heatmap, cv2.COLORMAP_JET)

# convert PIL image to numpy
img_np = np.array(img)

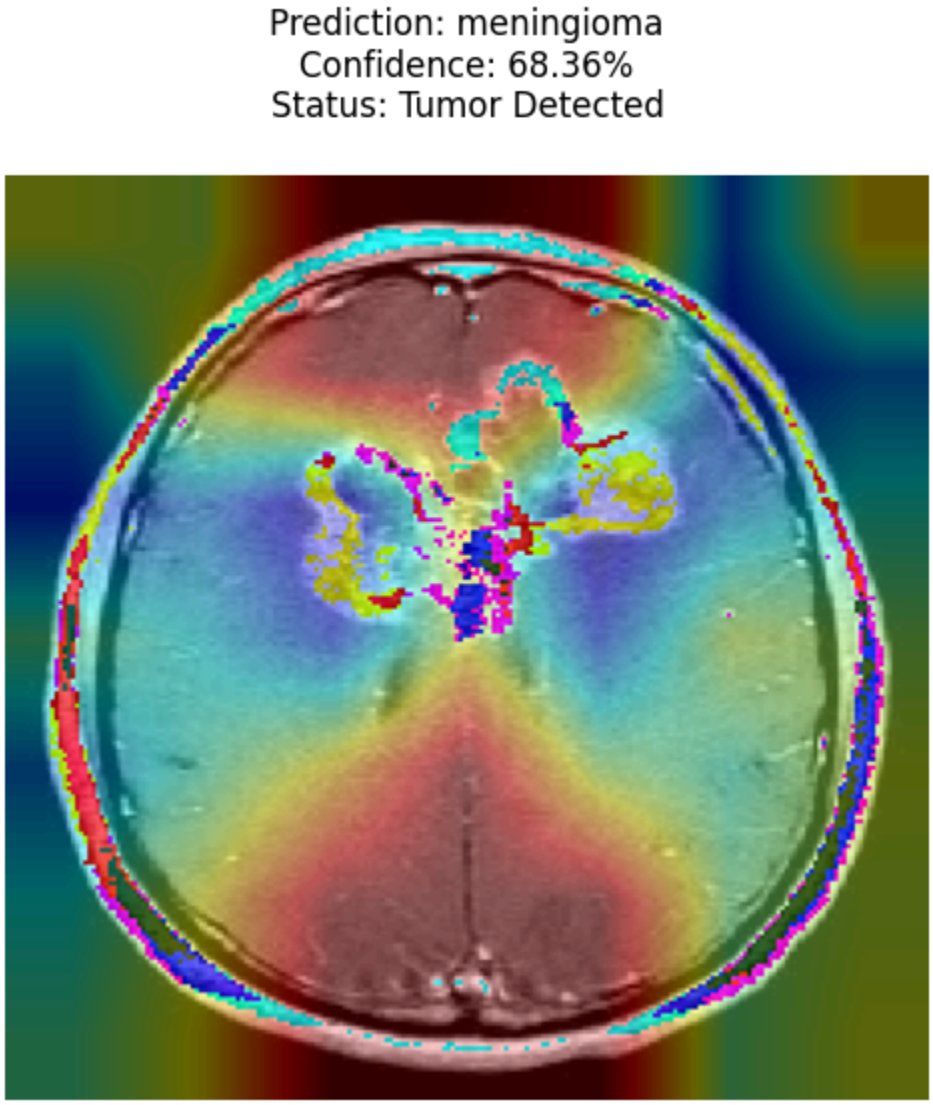
# overlay heatmap
superimposed_img = heatmap * 0.4 + img_np
superimposed_img = np.uint8(superimposed_img)

/usr/local/lib/python3.12/dist-packages/keras/src/models/functional.py:241: Us
Expected: [['keras_tensor']]
Received: inputs=Tensor(shape=(1, 224, 224, 3))
  warnings.warn(msg)


import matplotlib.pyplot as plt

plt.figure(figsize=(6,6))
plt.imshow(superimposed_img)
plt.title(f"""
```

```
Prediction: {predicted_label}
Confidence: {confidence*100:.2f}%
Status: {tumor_status}
""")
plt.axis("off")
plt.show()
```



png

```
# Measuring the Inference Time
import time

img_array = np.expand_dims(images[0], axis=0)

start = time.time()
model.predict(img_array)
end = time.time()
```

```
print("Inference Time (ms):", (end - start) * 1000)

◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 44ms/step
Inference Time (ms): 116.61505699157715

## installing lime
!pip install lime shap

Collecting lime
  Downloading lime-0.2.0.1.tar.gz (275 kB)
◇[?25l     ◇[90m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━◇[0m ◇[32m0.0/275.7
◇[?25h  Preparing metadata (setup.py) ... ◇[?25l◇[?25hdone
Requirement already satisfied: shap in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: scikit-learn>=0.18 in /usr/local/lib/python3.1
Requirement already satisfied: scikit-image>=0.12 in /usr/local/lib/python3.1
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: packaging>20.9 in /usr/local/lib/python3.12/di
Requirement already satisfied: slicer==0.0.8 in /usr/local/lib/python3.12/dis
Requirement already satisfied: numba in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: llvmlite in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: cloudpickle in /usr/local/lib/python3.12/dist-
Requirement already satisfied: typing-extensions in /usr/local/lib/python3.12/
Requirement already satisfied: networkx>=3.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: pillow>=10.1 in /usr/local/lib/python3.12/dist
Requirement already satisfied: imageio!=2.35.0,>=2.33 in /usr/local/lib/pytho
Requirement already satisfied: tifffile>=2022.8.12 in /usr/local/lib/python3.
Requirement already satisfied: lazy-loader>=0.4 in /usr/local/lib/python3.12/
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/di
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-pac
Building wheels for collected packages: lime
  Building wheel for lime (setup.py) ... ◇[?25l◇[?25hdone
  Created wheel for lime: filename=lime-0.2.0.1-py3-none-any.whl size=283834 s
  Stored in directory: /root/.cache/pip/wheels/e7/5d/0e/4b4fff9a47468fed56332
Successfully built lime
Installing collected packages: lime
```

```
Successfully installed lime-0.2.0.1

from lime import lime_image
from skimage.segmentation import mark_boundaries
import numpy as np
import matplotlib.pyplot as plt

## prediction functions for lime
def predict_fn(images):
    images = np.array(images)
    return model.predict(images)

explainer = lime_image.LimeImageExplainer()

explanation = explainer.explain_instance(
    img_array[0],
    predict_fn,
    top_labels=1,
    hide_color=0,
    num_samples=1000
)

  0%|          | 0/1000 [00:00<?, ?it/s]


◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m26s◇[0m 26s/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 57ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 54ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 52ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 51ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 54ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 79ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 71ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 70ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 69ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 69ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 66ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 67ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 79ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 78ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 64ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 71ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 74ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 72ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 72ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 66ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 53ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 54ms/step
◇[1m1/1◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m0s◇[0m 55ms/step
```

```
[1m1/1[0m [32m━━━
```

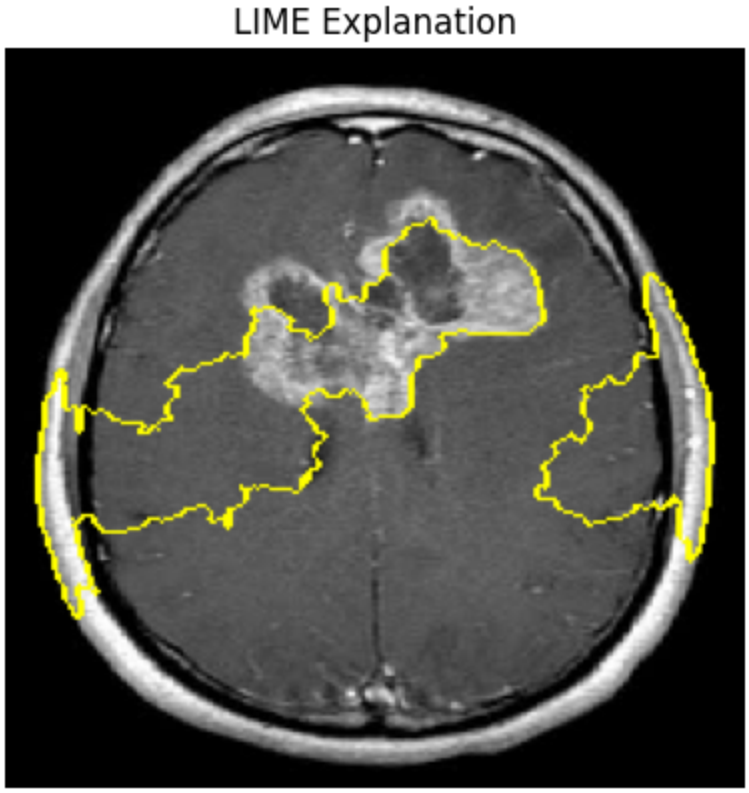
```
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 58ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 54ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 53ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 53ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 59ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 54ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 63ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 54ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 52ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 54ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 54ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 52ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 56ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 53ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 52ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 51ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 53ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 57ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 54ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 51ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 53ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 53ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 55ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 62ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 53ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 54ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 53ms/step
◆[1m1/1◆[0m ◆[32m━━━━━━━━━━━━━━━━━━━━◆[0m◆[37m◆[0m ◆[1m0s◆[0m 56ms/step
```

```
## Visualizing the superpixel part through lime
lime_img, mask = explanation.get_image_and_mask(
    explanation.top_labels[0],
    positive_only=True,
    num_features=5,
    hide_rest=False
)

lime_result = mark_boundaries(lime_img, mask)

plt.imshow(lime_result)
plt.title("LIME Explanation")
plt.axis("off")
```

```
(np.float64(-0.5), np.float64(223.5), np.float64(223.5), np.float64(-0.5))
```



png

```
## Applying SHAP
# installing SHAP

import shap

# creating image Masker which tells SHAP how to hide parts of the image.
masker = shap.maskers.Image("inpaint_telea", (224,224,3))

# SHAP explaner
explainer = shap.Explainer(model, masker)

shap_values = explainer(img_array)
```

```
  0%|          | 0/498 [00:00<?, ?it/s]

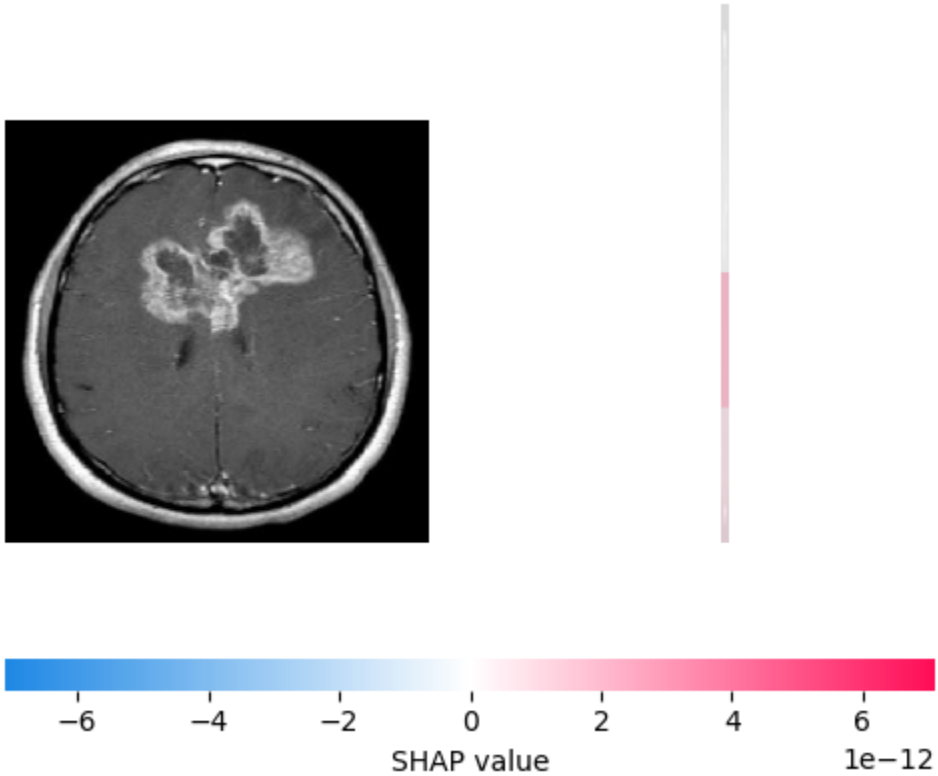

PartitionExplainer explainer: 2it [00:21, 21.89s/it]
```

```
## Visualixation of SHAP
# get prediction
pred = model.predict(img_array)
pred_class = np.argmax(pred)

# visualize shap
```

```
shap.image_plot(shap_values[0].values, img_array)
```

```
[1m1/1[0m [32m━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 41ms/step
```



png

```
plt.figure(figsize=(18,5))

# Original MRI
plt.subplot(1,4,1)
plt.imshow(img)
plt.title("Original MRI\nInput brain MRI used for tumor prediction")
plt.axis("off")

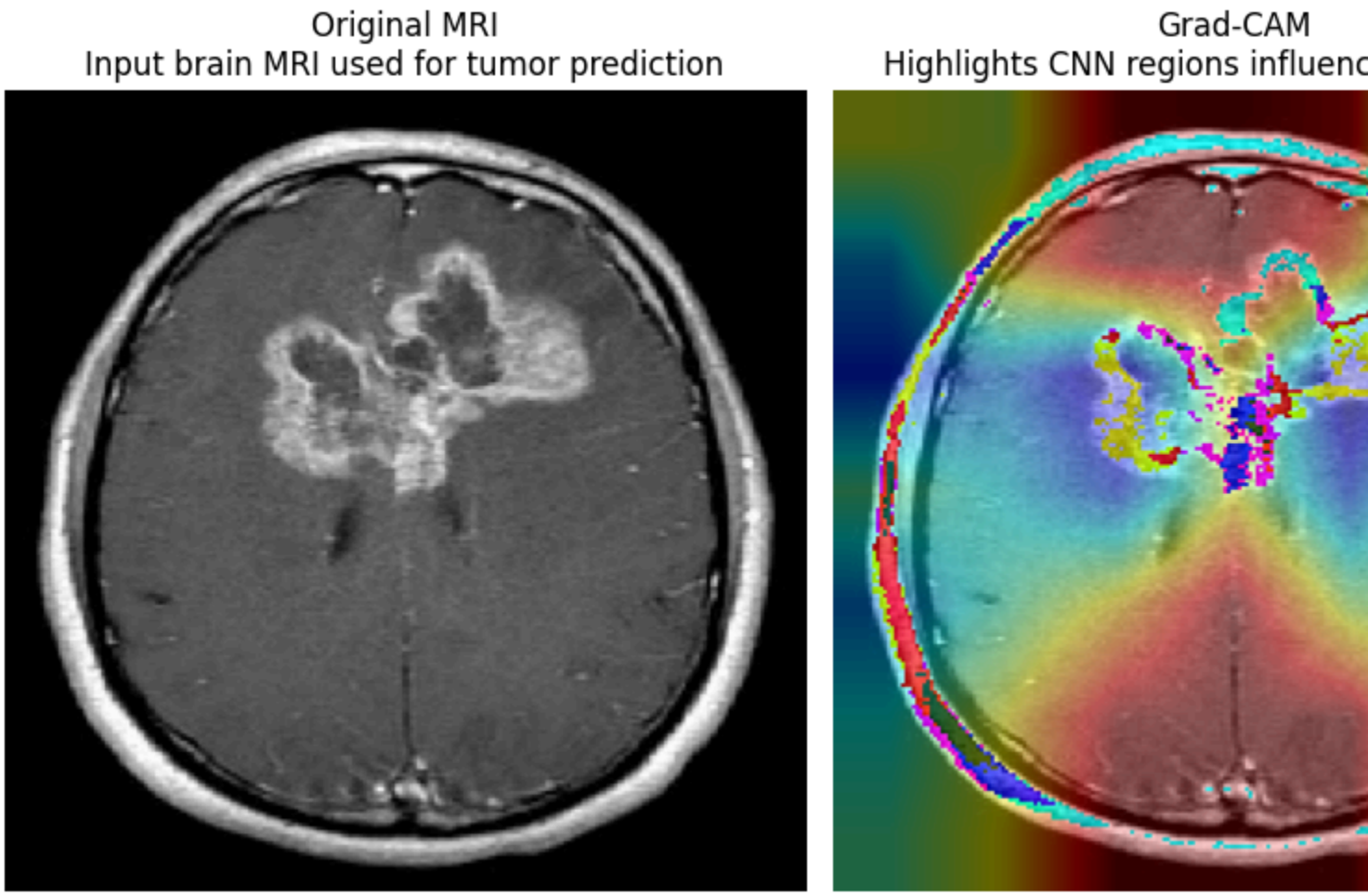
# Grad-CAM
plt.subplot(1,4,2)
plt.imshow(superimposed_img)
plt.title("Grad-CAM\nHighlights CNN regions influencing prediction")
plt.axis("off")

# LIME
plt.subplot(1,4,3)
plt.imshow(lime_result)
plt.title("LIME\nImportant superpixels affecting classification")
plt.axis("off")

# SHAP
```

```
plt.subplot(1,4,4)
plt.imshow(img)
plt.title("SHAP\nPixel-level contribution using Shapley values")
plt.axis("off")

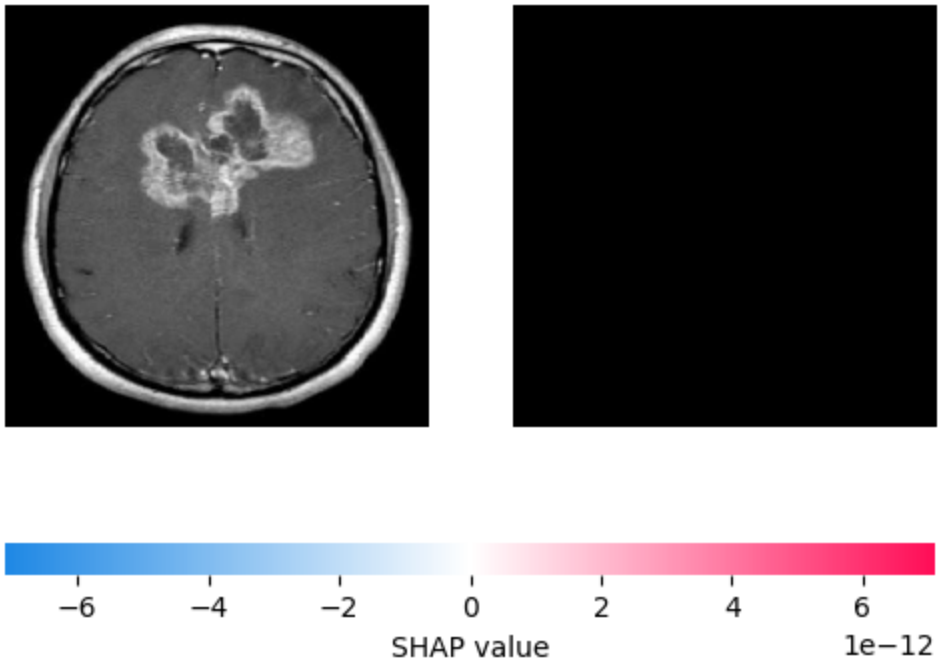
plt.tight_layout()
plt.show()
```



png

```
shap.image_plot(shap_values.values, img_array)
```

```
WARNING:matplotlib.image:Clipping input data to the valid range for imshow wi
```



png

```
import pandas as pd

comparison = pd.DataFrame({
    "Method": ["Grad-CAM","LIME","SHAP"],
    "Explanation Level": ["Region","Superpixel","Pixel"],
    "Speed": ["Fast","Medium","Slow"],
    "Interpretability": ["Medium","High","Very High"]
})

comparison
```

	Method	Explanation Level	Speed	Interpretability
0	Grad-CAM	Region	Fast	Medium
1	LIME	Superpixel	Medium	High
2	SHAP	Pixel	Slow	Very High

```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-05b63437-4a
        title="Convert this dataframe to an interactive table."
        style="display:none;">
```

```
<style>
  .colab-df-generate {
    background-color: #E8F0FE;
    border: none;
    border-radius: 50%;
    cursor: pointer;
    display: none;
    fill: #1967D2;
    height: 32px;
    padding: 0 0 0 0;
    width: 32px;
  }

  .colab-df-generate:hover {
    background-color: #E2EBFA;
    box-shadow: 0px 1px 2px rgba(60, 64, 67, 0.3), 0px 1px 3px 1px rgba(60, 64
    fill: #174EA6;
  }

  [theme=dark] .colab-df-generate {
    background-color: #3B4455;
    fill: #D2E3FC;
  }

  [theme=dark] .colab-df-generate:hover {
    background-color: #434B5C;
    box-shadow: 0px 1px 3px 1px rgba(0, 0, 0, 0.15);
```

```
    filter: drop-shadow(0px 1px 2px rgba(0, 0, 0, 0.3));
    fill: #FFFFFF;
  }
</style>
<button class="colab-df-generate" onclick="generateWithVariable('comparison')
        title="Generate code using this dataframe."
        style="display:none;">
```

```
</div>

import pandas as pd
import numpy as np

# get predictions for entire test set
pred = model.predict(test_generator)
pred_classes = np.argmax(pred, axis=1)

class_names = list(train_generator.class_indices.keys())

# create dataframe
sample_df = pd.DataFrame({
    "Image": [path.split("/")[-1] for path in test_generator.filepaths[:15]],
    "Actual Class": [class_names[i] for i in test_generator.classes[:15]],
    "Predicted Class": [class_names[i] for i in pred_classes[:15]],
    "Confidence": np.max(pred[:15], axis=1)
})

sample_df
```

```
◇[1m50/50◇[0m ◇[32m━━━━━━━━━━━━━━━━━━━━◇[0m◇[37m◇[0m ◇[1m3s◇[0m 60ms/step
```

	Image	Actual Class	Predicted Class	Confidence
0	Te-gl_1.jpg	glioma	meningioma	0.683645
1	Te-gl_10.jpg	glioma	glioma	0.977928
2	Te-gl_100.jpg	glioma	glioma	0.863377
3	Te-gl_101.jpg	glioma	pituitary	0.585024
4	Te-gl_102.jpg	glioma	pituitary	0.814457
5	Te-gl_103.jpg	glioma	meningioma	0.963094
6	Te-	glioma	meningioma	0.498187

	Image	Actual Class	Predicted Class	Confidence
	gl_104.jpg			
7	Te-gl_105.jpg	glioma	glioma	0.995691
8	Te-gl_106.jpg	glioma	meningioma	0.711070
9	Te-gl_107.jpg	glioma	glioma	0.823731
10	Te-gl_108.jpg	glioma	glioma	0.685930
11	Te-gl_109.jpg	glioma	pituitary	0.536274
12	Te-gl_11.jpg	glioma	glioma	0.967200
13	Te-gl_110.jpg	glioma	meningioma	0.843101
14	Te-gl_111.jpg	glioma	glioma	0.993205

```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-192114f6-88
        title="Convert this dataframe to an interactive table."
        style="display:none;">
```

```
<style>
  .colab-df-generate {
    background-color: #E8F0FE;
    border: none;
    border-radius: 50%;
    cursor: pointer;
    display: none;
    fill: #1967D2;
    height: 32px;
    padding: 0 0 0 0;
    width: 32px;
  }

  .colab-df-generate:hover {
    background-color: #E2EBFA;
    box-shadow: 0px 1px 2px rgba(60, 64, 67, 0.3), 0px 1px 3px 1px rgba(60, 64
    fill: #174EA6;
  }

  [theme=dark] .colab-df-generate {
    background-color: #3B4455;
    fill: #D2E3FC;
  }

  [theme=dark] .colab-df-generate:hover {
    background-color: #434B5C;
    box-shadow: 0px 1px 3px 1px rgba(0, 0, 0, 0.15);
    filter: drop-shadow(0px 1px 2px rgba(0, 0, 0, 0.3));
    fill: #FFFFFF;
  }
</style>
<button class="colab-df-generate" onclick="generateWithVariable('sample_df')"
        title="Generate code using this dataframe."
        style="display:none;">

🪄

</div>

import os
print(os.listdir())

['.config', 'sample_data']
```

```
import nbformat

file = "XAI222.ipynb"   # your notebook name

nb = nbformat.read(file, as_version=4)

# Remove widget metadata completely
nb.metadata.pop("widgets", None)

# Also clean inside cells (important)
for cell in nb.cells:
    if "metadata" in cell and "widgets" in cell["metadata"]:
        cell["metadata"].pop("widgets", None)

nbformat.write(nb, file)

print("✅ Fixed notebook successfully!")
```

```
---------------------------------------------------------------------------

AttributeError                            Traceback (most recent call last)

/usr/local/lib/python3.12/dist-packages/nbformat/__init__.py in read(fp, as_ve
    168     try:
--> 169         buf = fp.read()
    170     except AttributeError:


AttributeError: 'str' object has no attribute 'read'


During handling of the above exception, another exception occurred:


FileNotFoundError                         Traceback (most recent call last)

/tmp/ipykernel_1872/4270898628.py in <cell line: 0>()
      3 file = "XAI222.ipynb"   # your notebook name
      4
----> 5 nb = nbformat.read(file, as_version=4)
      6
      7 # Remove widget metadata completely


/usr/local/lib/python3.12/dist-packages/nbformat/__init__.py in read(fp, as_ve
    169         buf = fp.read()
    170     except AttributeError:
--> 171         with open(fp, encoding="utf8") as f:  # noqa: PTH123
```

```
    172             return reads(f.read(), as_version, capture_validation_erro
    173
```

```
FileNotFoundError: [Errno 2] No such file or directory: 'XAI222.ipynb'
```